MIL-T-24387A(SHIPS)
AMENDMENT-3
10 April 1979
SUPERSEDING
AMENDMENT-2
7 March 1977

MILITARY SPECIFICATION

TEMPERATURE MEASUREMENT EQUIPMENT

SIGNAL CONDITIONER AND POWER SUPPLY

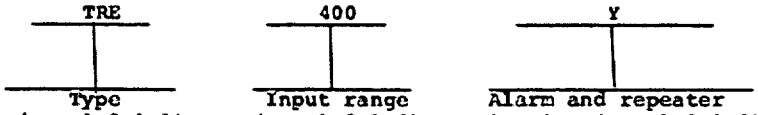
(ELECTRICAL) (NAVAL SHIPBOARD USE)

This amendment forms a part of Military Specification MIL-T-24387A(SHIPS), dated 1 June 1973 and is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

# 1.2.1, delete and substitute:

"1.2.1 The classification shall be of the following format, as specified (see 6.2.1):




# Add as paragraph 1.2.1.3:

"1.2.1.3 Alarm and repeater circuits. Temperature measurement equipment shall include an alarm circuit and a repeater circuit when needed to support system requirements (see 3.4.2.14 and 3.4.2.15). The presence of an alarm and repeater circuit shall be designated as follows:

Y - Yes - an alarm circuit and repeater circuit are included.
N - No - alarm and repeater circuits are not included."

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3.2(c): Delete "varified" and substitute "varied".

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# 3.2, add as items (e) and (f):

"(e) Actuation value. The difference between the temperature at which the alarm is actuated and the temperature at which the alarm is deactivated.
"(f) Alarm circuit repeatability. The ability of the alarm circuit to reproduce the actuation point and deactuation point."

# 3.4.2.2, second sentence: Delete and substitute: "The output shall be a true current source of 4 to 20 mA, d.c. The static error band requirement shall be met (see 3.10.1) regardless of external load resistance variations over a span of 0 to 250 ohms."

# 3.4.2.5, line 1: Delete "and span" and substitute "span and alarm".

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3.4.2.7, delete and substitute:

"3.4.2.7 Isolation. Input and output circuits shall be isolated from ground."

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Add as paragraphs 3.4.2.14 and 3.4.2.15:

"3.4.2.14 Alarm circuit. An alarm circuit shall be provided which will deenergize a relay when a set temperature is exceeded. The relay shall be wired to provide contact closures when the relay is deenergized thereby providing fail-safe operation (i.e., alarm sounds if circuit is inadvertently deactivated). The alarm circuit shall meet the following requirements when subjected to the test specified in 4.8.16:

(a) Set point accuracy - plus or minus 2 percent of span.
(b) Set point range of adjustment - 40 to 90 percent of span.
(c) Actuation value (see 3.2) - 2 percent of span.
(d) Alarm circuit repeatability (see 3.2) - 0.3 percent of span.
(e) Relay - The relay used in conjunction with alarm circuit shall be at least 2 pole double throw having a minimum continuous contact rating of 115 volts, 60 Hz, 5.0 amperes, resistive.

"3.4.2.15 Repeater circuit. The static error band (see 3.2) and repeatability of the repeater circuit shall be as specified in 3.10.1 for up to two additional indicators (repeaters) during the various conditions specified in 4.8.17. The circuit shall be so designed that the failure of the primary indicator or any of the repeaters or an open circuit to the primary indicator, or any of the repeaters shall not affect the others. Terminals or an A/N type connector shall be provided on the signal conditioner/power supply (SC/PS) to connect the repeaters."

3.5.1.1, add: "0° to 700°F".

3.5.2.1, add: "0° to 700°F".

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3.7.4, item (c) (2) d. 7 and 8: Delete "50°C" and substitute "65°C".

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3.10.9, line 2: Delete "0.04 percent per degree Fahrenheit" and substitute "0.02 percent per degree Celsius".

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3.10.13, line 2: Delete "between circuits and".

4.3, add: "Tests shall be conducted in the order listed in table I."

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Tables I and II, add the following:

"

| Examination and tests | Requirement paragraph | Test paragraph |
|---|---|---|
| Alarm circuit operation 1/<br>Repeater circuit operation 1/ | 3.4.2.14<br>3.4.2.15 | 4.8.16<br>4.8.17 |

1/ Simulate temperature element input."

Table III, add the following:

"

| Tests | Requirement paragraph | Test paragraph |
|---|---|---|
| Alarm circuit operation 1/<br>Repeater circuit operation 1/ | 3.4.2.14<br>3.4.2.15 | 4.8.16<br>4.8.17 |

1/ Simulate temperature element input."

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4.8.10.1.1 and 4.8.10.1.2, line 3: Delete "superimposed" and substitute "induced".

4.8.12, lines 2 and 3: Delete "and between these circuits".



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Add as paragraphs 4.8.16 and 4.8.17:

"4.8.16 Alarm circuit operation. The alarm circuit shall be tested for setpoint accuracy and repeatability by testing the indicator and alarm system at a temperature of 65 percent of span. The test temperature shall be approached three times and the alarm setpoint noted. Temperature shall be decreased after each run and the actuation value noted. The setpoint range of adjustment shall be tested at 40 percent and 90 percent of span. Parameters shall be in accordance with 3.10.1.

"4.8.17 Repeater circuit operation. Two electrical repeaters shall be connected to the indicating system and the following tests conducted with a reference measurement in accordance with 4.8.2 taken at each condition. Short circuit conditions are to be simulated with a shorting bar in place of the repeater."

NOTE: The margins of this amendment are marked "#" to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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